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Theoretical Model of the KrF Laser in the Direct Discharge

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Abstract:

In this work the main reactions governing the action of the KrF-laser are discussed. Rate constants of the electronic processes are calculated by approximating the electron distribution by a Maxwellian with an effective temperature T_e. The pumping energy for the electrons is determined by the experimentally observed voltage and current. The discussion of the reaction kinetics shows the vanishing influence of any dimerization of the Kr-species. The most important processes, i.e. the formation and deactivation of KrF* occur on the same time scale. The life time of KrF* amounts to about 2 ns in typical gas mixtures. In the present model the laser radiation is described in terms of an ordinary rate equation and the laser transition is considered to be homogeneously broadened. Inhomogenities of the pump power in the laser gas are neglected. A computer study shows that from a total of 31 parameters entering the model only four have a critical influence on the results.

These are the:

(a) electron attachment rate of the F_2-molecule

(b) electron temperature, or the electron distribution

(c) the spontaneous lifetime and the cross section for stimulated emission and

(d) distributed losses inside the cavity.

Thus the calculated results show an error of about 50%. A comparison with experiments yields deviations in the efficiency by at least this factor. By adjusting the electron temperature and the distributed losses inside the cavity good agreement between experiment and model can be achieved.

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A Introduction

The noble gas halide lasers, discovered in 1975, have become the most common laser light source in the ultraviolet spectral range in recent years. The possibility of exciting these lasers in a direct gas discharge – similar to the nitrogen laser – along with the wide range of laser lines (see Fig. 1), has contributed to the rapid development of the RGH laser (Rare Gas Halide laser). Other important advantages include the relatively high efficiencies achieved (1–2%) and, although limited, some tunability. A comprehensive overview of this laser group can be found in [1].

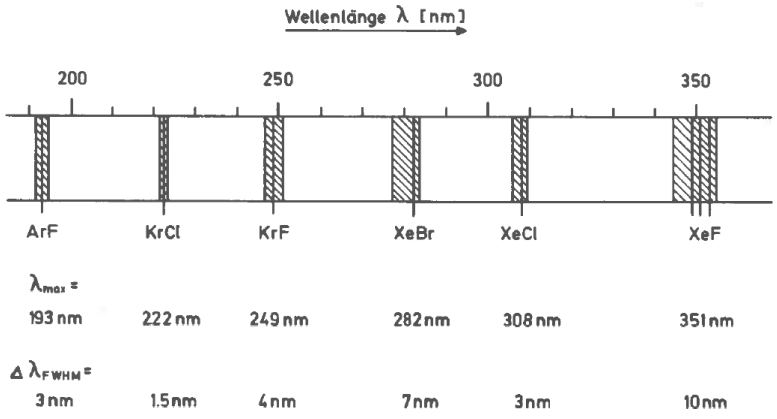


Figure 1: Overview of the emission wavelengths $\lambda_{\max}$ of noble gas halide lasers. $\Delta\lambda_{\mathrm{FWHM}}$ represents the half-width of the fluorescence of the various excimers.

In general, the fact that electronically excited noble gas atoms are nearly isoelectronic with the corresponding alkali atoms is exploited in RGH lasers. They react very vigorously with halogens, forming noble gas halides (see Fig. 2). These molecules, which are bound only in the excited state (excimers), have a short lifetime of a few ns. They decay (i.e., dissociate) by emitting a photon. The essential pumping process in these lasers is the excitation of the noble gas atoms, which can occur through electron impact in an electron beam setup or in a discharge. Excitation in a discharge offers practical advantages, as these devices are easy to handle and the repetition frequency, and thus the average power, is much higher than in corresponding electron beam setups.

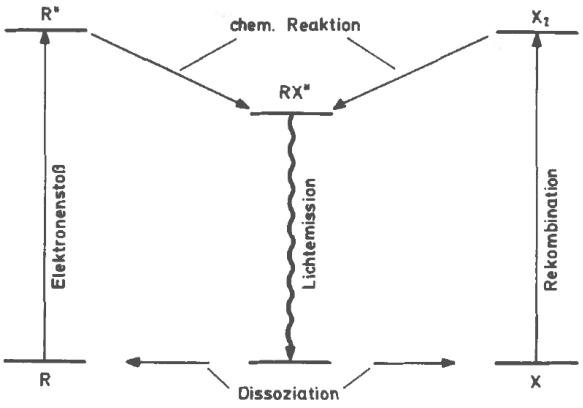


Figure 2: Pumping scheme of the noble gas halide lasers. R = noble gas atom, X = halogen atom

The most powerful laser in the RGH laser group is the KrF laser, which, when excited in a direct discharge, delivers up to 1 joule of energy in 30 ns [2]. Theoretical models for this laser were primarily developed at AVCO [3], which only deal with electron beam-pumped systems. A more detailed calculation of the processes occurring in a discharge was carried out by Greene and C. Brau [4]. Starting from the electrical discharge circuit and taking into account the electron distribution obtained with a Boltzmann code, the KrF laser was simulated.

The goal of this work is to identify and discuss the key reactions and processes in the KrF laser. Based on current and voltage measurements of a KrF laser, the balance equations for the various particles in the discharge are established. The light field is described using the rate equation model, and finally, the results of this model are compared with experimental data. The KrF laser serves here as a model for the other RGH lasers.

B Model

B.1 Electron balance

B.1.a Description of the discharge

Typical laser gas mixtures have a composition of approximately 85–90% helium, 10–15% krypton, and about 0.3–0.6% fluorine. During the pre-ionization phase, electrons are released by UV light [5]. These electrons are then accelerated by the applied voltage until, in a collision, they release at least part of their energy. Depending on the electron energy, the following collision processes are distinguished:

(a) Elastic collisions, where kinetic energy is transferred, and

(b) Inelastic collisions, where excitation, dissociation, or ionization of molecules occurs.

The characteristic quantity for all these collision processes is the collision cross section. Fig. 3 shows the elastic collision cross sections of the noble gases. In contrast to He and Ne, Ar, Kr, and Xe show the typical Ramsauer minima at an electron energy of 1 eV (for a discussion of the collision processes, see e.g., [6]). For excitation by collisions, approximately 10 eV are required for Kr (see Fig. 4). This means that the electron temperature in the discharge must be at least a few eV to ensure that enough fast electrons are available for excitation.

For a qualitative understanding of the discharge, we will briefly explain how the collision cross section σ affects the resistance of a discharge, and thus also determines the energy coupled into the gas (and, consequently, the electron temperature). For simplicity, it will be assumed that σ is independent of the electron energy.

The current density $\vec{j}$ in a discharge is given by

$$\vec{j} = s\vec{E}$$

where s is the conductivity per cm and $\vec{E}$ is the electric field strength.

The drift velocity $\vec{v}_D$ of the electron cloud in the gas is defined by:

$$\vec{j} = en_e\vec{v}_D \tag{B1-1}$$

where e is the elementary charge and n_e is the electron density.

The higher the electric field, the higher the drift velocity. The proportionality constant is referred to as the mobility k:

$$\vec{v}_D = k\vec{E}. \tag{B1-2}$$

On the other hand, the drift velocity $\vec{v}_D$ can be estimated using the simplest model of Langevin [6]:

$$\vec{v}_D = \frac{e\lambda}{m\bar{v}}\vec{E} \tag{B1-3}$$

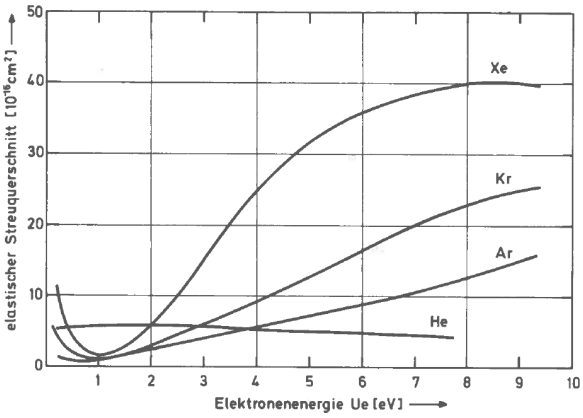


Figure 3: Elastic scattering cross section of electrons on various noble gases according to [7a].

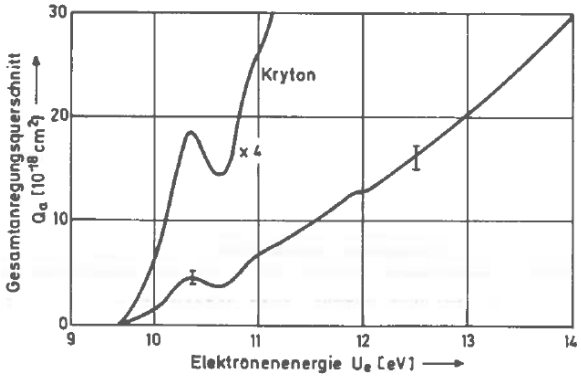


Figure 4: Excitation cross section of krypton for electron collisions according to [7b].

where λ is the mean free path of the electrons

$$\lambda = \frac{1}{N\sigma} \tag{B1-4}$$

where N is the neutral particle density and σ is the elastic scattering cross section, and

$$\bar{v} = 6.7 \times 10^7 \sqrt{T_{\mathrm{ev}}} \tag{B1-5}$$

is the mean electron velocity at an electron temperature T_{ev}.

Thus, we get:

$$\begin{aligned} \vec{\mathrm{v}}_{\mathrm{D}} &= \frac{e\vec{\mathrm{E}}}{mN\sigma 6.7 \times 10^7 \sqrt{T_{\mathrm{ev}}}} \\ k &= \frac{e}{mN\sigma 6.7 \times 10^7 \sqrt{T_{\mathrm{ev}}}} \\ s &= \frac{e^2 n_{\mathrm{e}}}{mN\sigma 6.7 \times 10^7 \sqrt{T_{\mathrm{ev}}}}. \end{aligned} \tag{B1-6}$$

These formulas of course give only rough estimates, but they show the key relationships for a qualitative understanding:

(a) For the same electron density n_{e}, the current remains the same if the quotient $\vec{\mathrm{E}}/N$ remains constant. Therefore, at higher gas pressures, a higher $\vec{\mathrm{E}}$ must be chosen.

(b) Small scattering cross sections result in high drift velocities and thus a small resistance of the discharge (low energy coupling).

(c) An increase in the density results in a higher resistance.

(d) High electron energies are desirable for a high discharge resistance.

A precise calculation of the current and the coupled energy must be based on the energy-dependent σ values. Typically, a Maxwell distribution of the electrons is not established. Instead, it must be obtained by solving the Boltzmann equation. From this, the drift velocity $\vec{\mathrm{v}}_{\mathrm{D}}$ or mobility k can then be determined [8].

In the model used here, the discharge quantities, such as current and voltage, are not calculated. They are measured in the experiment and entered into the program. That is, from the energy coupled into the gas, the electron energy and excitation rates are calculated. We assume that the electron gas is heated by an external voltage source. In this heating process, the electrons lose energy both due to thermalization and as a result of excitation or ionization collisions (elastic and inelastic collisions). The excited or ionized atoms then react further to form exciplexes, which are ultimately responsible for the laser emission. The relationship in our model is illustrated in Fig. 5.

The arrows ↑↓ next to T_{e} and n_{e} indicate an increase or decrease in the respective process.

According to this model, the electron density and the electron temperature are described by two equations:

1. Equation for electron energy:

$$\frac{\mathrm{d}\left(\frac{3}{2} n_{\mathrm{e}} kT\right)}{\mathrm{d}t} = \text{Energy input} - \text{Energy loss through elastic collisions} - \text{Energy loss through inelastic collisions.} \tag{B1-7}$$

2. Equation for electron density:

$$\frac{\mathrm{d}n_{\mathrm{e}}}{\mathrm{d}t} = \text{Increase through ionization collisions} - \text{Decrease through electron attachment.} \tag{B1-8}$$

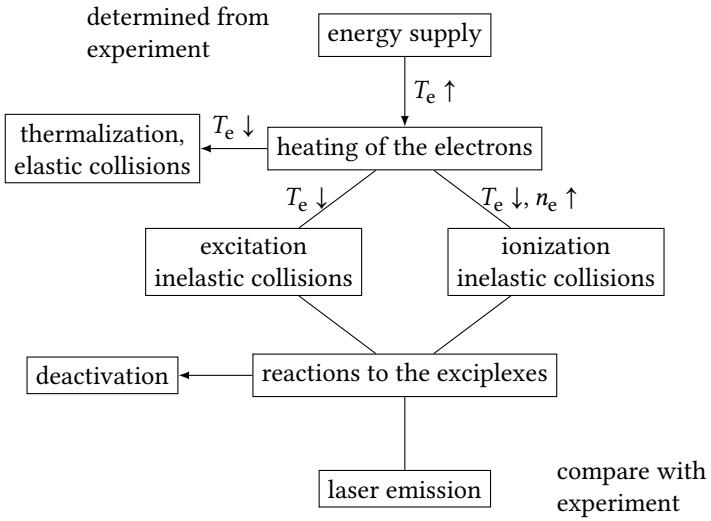


Figure 5: Schematic flow of the program

The last term in equation (B1-8) describes the electron losses due to the attachment of slow electrons to F_2 molecules. This process will be discussed in more detail later.

In the further course of this chapter, these two equations will be elaborated, and the individual collision processes will be discussed.

B.1.b Elastic collisions

α) **e–e collisions** First, the question arises as to how quickly the electrons relax with each other. The characteristic time τ_{ee} for this process is given by [9]:

$$\tau_{ee} = \frac{(T_e^0)^{3/2}}{3.8 n_e \ln \Lambda}.$$

At the typical temperatures of discharge plasmas, which are in the range of a few eV, $\ln \Lambda \approx 10$, and it follows that:

$$\tau_{ee} = 3.3 \times 10^4 \, \frac{(T_e^0)^{3/2}}{n_e},$$

that is, for $n_e = 10^{16}$ and $T_e = 5\,\text{eV}$, the relaxation time τ_{ee} is about 40 ps; thus, the electrons are in equilibrium with each other.

β) **e–He collisions** The mean time between two collisions is:

$$\tau_{e\text{-He}} = (\sigma_{e\text{-He}} n_{He} \bar{v}_e)^{-1}, \tag{B1-9}$$

where n_{He} is the helium particle density and $\bar{v}_e$ is the mean thermal velocity of the electrons. At 1000 torr He and $\sigma_{e\text{-He}} \approx 5 \times 10^{-16}\,\text{cm}^2$, the value of $\tau_{e\text{-He}}$ is approximately 4×10^{-13} sec. In each collision, the maximum energy that can be transferred is in the ratio of the masses of the particles: $\Delta E_{transfer} = \frac{2m_e}{M_{He}} E_e$.

Thus, a large number $\frac{M_{He}}{2m_e}$ of collisions are needed for energy equilibrium. More precisely, it holds [9]:

$$\frac{\mathrm{d}T_\mathrm{e}}{\mathrm{d}t} = -\frac{T_\mathrm{e} - T_0}{\tau_{\text{e-He}}(M_\mathrm{He}/2m_\mathrm{e})} = \frac{T_\mathrm{e} - T_0}{t_{\text{e-He}}}$$

and the temperature difference between the electrons and helium decreases by the factor $1/e$ within about 1.5 ns. The temperature T_0 of the helium changes very little due to the discharge. The total energy supplied to the gas is typically 6×10^{-2} joule/cm^3. This results in a maximum temperature increase of about 80 K, i.e., helium acts like a heat bath.

γ) **e–Kr collisions** Due to the low krypton density (about 1/10th of the helium density) and the much higher mass ratios, the corresponding times for e–Kr collisions are much larger. Therefore, elastic e–Kr collisions can be neglected in the energy balance.

Finally, the interesting question arises as to what temperature the electrons would reach if only elastic collisions were considered. This is given by:

$$\frac{\mathrm{d}T_\mathrm{e}}{\mathrm{d}t} = +\frac{\epsilon}{\frac{3}{2}n_\mathrm{e}} - \frac{T_\mathrm{e}}{t_{\text{e-He}}}.$$

The first term here describes the external energy input. Assuming a discharge duration of 40 ns, the energy is $\epsilon = 1.5 \times 10^6$ joule/cm^3/s, meaning that with continuous energy supply, an equilibrium value will be reached at electron densities of 10^{16}/cm^3 at T_ev (electron) ≈ 1 eV. Only at the beginning of the discharge, when the electron density is still low, will the electron temperature be significantly higher. From this discussion, the elastic collisions and electron heating can be described in equation (B1-7) by:

$$\frac{\mathrm{d}\left(\frac{3}{2}n_\mathrm{e}T_\mathrm{e}\right)}{\mathrm{d}t} = \epsilon(t) - \frac{\frac{3}{2}n_\mathrm{e}T_\mathrm{e}}{t_{\text{e-He}}} - (\text{terms of inelastic scattering}). \tag{B1-10}$$

$\epsilon(t)$ here is a measured quantity.

B.1.c Inelastic collisions

In the case of inelastic collisions, a distinction must be made between excitation and ionization collisions. Of the species present in the laser mixture, He, Kr, and F_2, only Kr and F_2 need to be discussed, since He, with an excitation potential of 19 eV, hardly contributes to the electron energy balance ($T_\mathrm{e} = 1$ eV). In Fig. 4, the excitation cross section for electron–krypton collisions is shown. The first excitation potential of krypton is 10 eV. Other excitation thresholds are at 10.6 eV and 12 eV, with the curvature of the curve indicating that different levels are excited in the intermediate range. In Fig. 6a, the energy level diagram of krypton atoms is shown [10]. In the following discussion, several levels will be simplified and grouped together as Kr* and Kr** as indicated in Fig. 6b. Kr* has an excitation potential of 10 eV, Kr** of 10.6 eV, and the ionization threshold of krypton is at 14 eV. The excitation cross sections for F_2 are not known. In the following discussion, measured rates will therefore be used for e–F_2 collisions (see below).

α) **Calculation of rate constants from cross sections** In general, the following relationship holds between cross section and rate constant: per dt, $(\sigma v n_1 n_2)$ reactive collisions occur between particle types n_1 and n_2, if their cross section is σ and v is the velocity of the particles, i.e., for constant particle velocity, the rate constant is $K = \sigma v$, where the rate constant is given in cm^3/sec. If the electrons have an energy distribution $f(\epsilon)$, the excitation or ionization rates can be obtained from the cross sections by the following calculation:

$$n_\mathrm{e} = n_{\mathrm{e}0}\, f(\epsilon)\, \mathrm{d}\epsilon \qquad (\text{number of electrons in the energy range } \epsilon \text{ to } \epsilon + \mathrm{d}\epsilon).$$

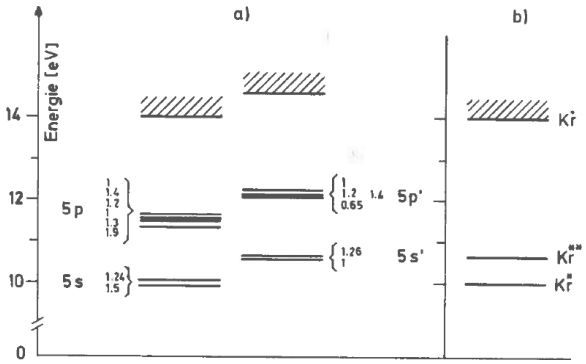


Figure 6: a) Energies of the various states of Kr according to the tables of C. E. Moore [10]. The first excited states 5s, 5s′, 5p and 5p′ are listed, along with their corresponding degeneracy factors. b) For simplification in the calculations, these states are grouped together as Kr* and Kr**.

With their cross section $\sigma(\epsilon)$ and their velocity $v(\epsilon)$, they fill the volume in dt: d$V = n_e v(\epsilon)\sigma(\epsilon)\,\mathrm{d}t$. In d$V$, however, there are $n_i\,\mathrm{d}V$ particles that will collide in dt. The total number of collisions A in dt can be obtained by integrating over the energy range:

$$A = \int_0^\infty n_e v(\epsilon)\sigma(\epsilon) n_i\,\mathrm{d}t = n_{e0} n_i \int_0^\infty v(\epsilon)\sigma(\epsilon) f(\epsilon)\,\mathrm{d}\epsilon$$

where $f(\epsilon)$ describes the energy distribution of the electrons. The reaction constant K is thus given by:

$$K = \int_0^\infty v\sigma f(\epsilon)\,\mathrm{d}\epsilon\,. \tag{B1-11}$$

If $\sigma(\epsilon)$ can be linearized near the excitation potential: $\sigma = \alpha(E - E_a)$ and $f(\epsilon)$ is the Maxwell energy distribution:

$$f(\epsilon) = \frac{2}{\pi^{1/2}} \frac{E^{1/2}}{(kT)^{3/2}} \exp\left(-\frac{E}{kT}\right) \mathrm{d}E\,,$$

then

$$K_{ei} = \left(\frac{8}{\pi m}\right)^{1/2} \alpha (kT)^{1/2} (E_a + kT) \exp\left(-\frac{E_a}{kT}\right). \tag{B1-12}$$

If we further denote by $\sigma_e^* = \alpha[(E_a + kT) - kT] = \alpha kT$ the excitation cross section at $(E_a + kT)$, we obtain, since $\bar{v}_e = (8kT/\pi m)^{1/2}$ is the average electron velocity, for K_{ei}:

$$K_{ei} = \bar{v}_e \sigma_{ei}^* \left(\frac{E_a}{kT} + 2\right) \exp\left(-\frac{E_a}{kT}\right). \tag{B1-13}$$

In the typical case, $f(\epsilon)$ will not follow a Maxwell distribution but will adjust for each E/N value in the discharge. In this case, K_{ei} must be calculated separately for each E/N using equation (B1-11).

The real electron distribution is obtained from the solution of the Boltzmann equation. However, in our further discussion, we assume that $f(\epsilon)$ is approximately described by the Maxwell distribution.

β) **Electron–krypton collisions** The e–Kr collisions can be divided into collisions with krypton atoms in the ground state and those with krypton atoms in excited states. In the following, we will discuss the collisions in the order of Kr, Kr*, and Kr**. The excitation potentials and the slope α of the excitation cross sections are indicated after the reaction equations [11].

$$\mathrm{Kr} + \mathrm{e} \rightarrow \mathrm{Kr}^* + \mathrm{e} \qquad E_\mathrm{a}^1 = 10\,\mathrm{eV},\ \alpha_1 = 11 \times 10^{-18}\,\mathrm{cm}^2/\mathrm{eV}, \tag{1}$$

$$\mathrm{Kr} + \mathrm{e} \rightarrow \mathrm{Kr}^{**} + \mathrm{e} \qquad E_\mathrm{a}^2 = 10.6\,\mathrm{eV},\ \alpha_2 = 2 \times 10^{-17}\,\mathrm{cm}^2/\mathrm{eV}, \tag{2}$$

$$\mathrm{Kr} + \mathrm{e} \rightarrow \mathrm{Kr}^{+} + \mathrm{e} \qquad E_\mathrm{a}^3 = 14.0\,\mathrm{eV},\ \alpha_3 = 2.6 \times 10^{-17}\,\mathrm{cm}^2/\mathrm{eV}. \tag{3}$$

The cross sections refer to the excitation processes, i.e., to the arrows pointing to the right. Of course, each reaction also has the corresponding reverse reaction, which generally occurs more rapidly than the excitation reaction, as the following consideration shows:

At a certain temperature T_e, for example:

$$\frac{\mathrm{dKr}^*}{\mathrm{d}t} = K(\mathrm{e}, \mathrm{Kr}, \mathrm{Kr}^*) n_\mathrm{e}\mathrm{Kr} - K(\mathrm{e}, \mathrm{Kr}^*, \mathrm{Kr}) n_\mathrm{e}\mathrm{Kr}^*,$$

where Kr or more generally A represents the particle density of Kr or A, respectively. K is the reaction constant, where the notation is such that the rate for the process $\mathrm{A} + \mathrm{B} \rightarrow \mathrm{C}$ is denoted by $K(\mathrm{A}, \mathrm{B}, \mathrm{C})$.

In thermal equilibrium, $\mathrm{dKr}^*/\mathrm{d}t = 0$ and $\frac{\mathrm{Kr}^*}{\mathrm{Kr}} = \frac{g^*}{g} \exp\left(-\frac{E_\mathrm{a}}{kT}\right)$.

This leads to:

$$\frac{K(\mathrm{e}, \mathrm{Kr}, \mathrm{Kr}^*)}{K(\mathrm{e}, \mathrm{Kr}^*, \mathrm{Kr})} = \frac{g^*}{g} \exp\left(-\frac{E_\mathrm{a}}{kT}\right),$$

and the deactivation rate is, except in cases of very high degeneracy of the upper state, always greater than the excitation rate. In general, a reaction direction can always be neglected if the relaxation time of the equilibrium is much longer than the process being considered. The relaxation time for equilibrium for (1) is [9]:

$$\tau = \frac{1}{\bar{v}_\mathrm{e} \sigma^* n_\mathrm{e}} \approx 300\,\mathrm{ns} \quad (\mathrm{at}\ T_\mathrm{e} = 1\,\mathrm{eV}),$$

where σ^* denotes the cross section at $(E_\mathrm{a} + kT)$. Similar considerations apply to reactions (2) and (3). In Fig. 7, the characteristic times for equilibrium relaxation are written next to the various arrows. This means that only the forward reactions in (1)–(3) need to be considered.

Collisions with excited krypton atoms are

$$\mathrm{Kr}^* + \mathrm{e} \rightarrow \mathrm{Kr}^{**} + \mathrm{e} \qquad E_\mathrm{a}^4 = 1.5\,\mathrm{eV},\ \alpha_4 = 7.5 \times 10^{-15}\,\mathrm{cm}^2/\mathrm{eV}, \tag{4}$$

$$\mathrm{Kr}^* + \mathrm{e} \rightarrow \mathrm{Kr}^{+} + 2\mathrm{e} \qquad E_\mathrm{i}^5 = 4.5\,\mathrm{eV},\ \alpha_5 = 1.7 \times 10^{-16}\,\mathrm{cm}^2/\mathrm{eV}, \tag{5}$$

$$\mathrm{Kr}^{**} + \mathrm{e} \rightarrow \mathrm{Kr}^{+} + 2\mathrm{e} \qquad E_\mathrm{i}^6 = 3\,\mathrm{eV},\ \alpha_6 = 1.7 \times 10^{-16}\,\mathrm{cm}^2/\mathrm{eV}. \tag{6}$$

For the reaction rates of Kr*, the corresponding values for rubidium are used [12]. The reaction constant for (4) at $T_\mathrm{e} = 1\,\mathrm{eV}$ is very high,

$$K(\mathrm{e}, \mathrm{Kr}^*, \mathrm{Kr}^{**}) = \sigma\bar{v}\left(\frac{E_\mathrm{a}}{kT} + 2\right) e^{-\frac{E_\mathrm{a}}{kT}} \approx 5 \times 10^{-7}\,\mathrm{cm}^3/\mathrm{s}.$$

Since the reverse reaction happens even faster: $K(\mathrm{e}, \mathrm{Kr}^{**}, \mathrm{Kr}^{*}) \approx 7.5 \times 10^{-7}\,\mathrm{cm}^3/\mathrm{s}$ [9], the timescale on which an equilibrium between Kr* and Kr** is reached is approximately given by: $\tau = (7.5 \times 10^{-7} n_\mathrm{e})^{-1} \approx 0.15\,\mathrm{ns}$ at $n_\mathrm{e} = 10^{16}$, which means that Kr* and Kr** are in thermal equilibrium. Thus, in the following discussion, we assume this equilibrium.

The ionization collisions (5) and (6) are significant for the particle balance of the electrons. For reaction (6), no values are available from the literature. Therefore, we have initially assumed that the slope of the cross section for this process is the same as for process (5). Due to the lower ionization potential of Kr**, process (5) will be the dominant ionization process. An indication of the relaxation time to equilibrium is given by $\tau = [K(\mathrm{e}, \mathrm{Kr}^{**}, \mathrm{Kr}^{+}) n_\mathrm{e}]^{-1} \approx 10^{-8}$ sec.

In Fig. 7, the relevant krypton levels are shown again along with the characteristic times. For this, densities for n_e, Kr, Kr* and Kr** of $10^{16}\,\mathrm{cm}^{-3}$ and temperatures of $T_\mathrm{e} \approx 1\,\mathrm{eV}$ were assumed. For time durations that are long compared to the discharge time (~ 40 ns), only the forward reactions need to be considered in our model, not the reverse reactions.

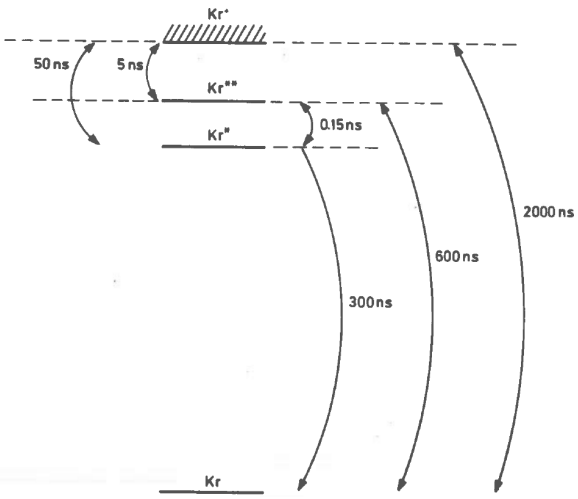


Figure 7:

γ) **Electron–fluorine collisions** For the electron density in the laser gas, the attachment of electrons to F_2 is particularly important, as this is the only reaction where electrons "disappear":

$$\mathrm{e} + \mathrm{F}_2 \rightarrow \mathrm{F}^- + \mathrm{F} \qquad K = 2 \times 10^{-9}\,\mathrm{cm}^3/\mathrm{sec}\ \text{at}\ 1\ \mathrm{eV}. \tag{7}$$

At temperatures of 1 eV, H. Chen et al. [13] measured a value for the reaction constant K of $2 \times 10^{-9}\,\mathrm{cm}^3/\mathrm{sec}$. The exothermicity of this reaction is 3.6 eV. As the temperature increases, K will decrease. Typical times for electron attachment at 1 eV are:

$$\tau_{\mathrm{e},\mathrm{F}_2} \approx 3\,\mathrm{ns}.$$

For the temperature dependence of the attachment rate, an exponential law $e^{-\alpha T_\mathrm{e}}$ is assumed, where the values for $T_\mathrm{e} = 0$ ($K(7) = 3 \times 10^{-9}$) and $T_\mathrm{e} = 1$ ($K(7) = 2 \times 10^{-9}$) are fitted:

$$K(\mathrm{e}, \mathrm{F}_2, \mathrm{F}) = 3 \times 10^{-9} \exp(-0.41 T_\mathrm{e})\,\mathrm{cm}^3/\mathrm{sec}.$$

δ) **Dimer recombination** Another attachment of electrons can occur to Kr_2^+ molecules, which are formed in small amounts during the discharge. These should only be mentioned here for completeness. In [14], for the reaction:

$$e + Kr_2^+ \rightarrow Kr^{**} + Kr \tag{8}$$

the following rate constant is given:

$$K(e, Kr_2^+, Kr) = 2.4 \times 10^{-7} (T_e)^{-1/2}.$$

B.1.d Electron balance equations

Based on the previous discussion, the following equation for the electron density n_e can be written:

$$\frac{dn_e}{dt} = K(e, Kr, Kr^+) n_e Kr + K(e, Kr^*, Kr^+) n_e Kr^* + K(e, Kr^{**}, Kr^+) n_e Kr^{**} - K(e, F_2, F) n_e F_2 - K(e, Kr_2^+, Kr) n_e Kr_2^+. \quad \text{(B1-14)}$$

The initial condition for the integration is determined by the electron density at the beginning of the discharge, which is assumed to be about $10^{10}\ cm^{-3}$: $n_e(t = 0) = 10^{10}\ cm^{-3}$.

This equation can be simplified, as Kr^* and Kr^{**} are in thermal equilibrium:

$$\frac{Kr^{**}}{Kr^*} = \underbrace{\frac{g^{**}}{g^*} \exp \frac{-\Delta E}{T_e}}_{a(T_e)}; \qquad E = 1.5\ eV; g^* = 2; g^{**} = 3.$$

One can express them using a single state:

$$Kr^* + Kr^{**} = \overline{Kr^*}$$

$$Kr^* = \frac{\overline{Kr^*}}{1 + a(T_e)}; \qquad Kr^{**} = \overline{Kr^*} \frac{a(T_e)}{1 + a(T_e)}$$

Thus, the two ionization terms can be combined:

$$n_e \left[K(e, Kr^*, Kr^+) Kr^* (1 + a(T_e))^{-1} + K(e, Kr^{**}, Kr^+) Kr^* \frac{a(T_e)}{1 + a(T_e)} \right] = K(e, Kr^*, Kr^+) n_e Kr^*$$

with

$$K(e, Kr^*, Kr^+) = \frac{1}{1 + a(T_e)} \left(K(e, Kr^*, Kr^+) + a(T_e) K(e, Kr^{**}, Kr^+) \right)$$

and (B1-14) becomes:

$$\frac{dn_e}{dt} = K(e, Kr, Kr^+) n_e Kr + K(e, \overline{Kr^*}, Kr^+) n_e \overline{Kr^*} - K(e, F_2, F) n_e F_2 - K(e, Kr_2^+, Kr) n_e Kr_2^+ \tag{B1-15}$$

The calculation of the K values is done using equation (B1-12), where the quantities α and E_a are given for the corresponding reactions (1)–(7).

The temperature is obtained from the particle energy balance (B1-10). It must be taken into account that during each ionization collision, additional energy $\frac{3}{2} T_e$ is lost, since the freed electron must also be heated to T_e. In the case of a recombination reaction, the electron gas loses energy, specifically $\frac{3}{2} T_e$. From equation (B1-10), this leads to:

$$
\begin{aligned}
\frac{\mathrm{d}\left(\frac{3}{2}n_\mathrm{e}T_\mathrm{e}\right)}{\mathrm{d}t} &= \epsilon(t) - \frac{3}{2}n_\mathrm{e}T_\mathrm{e}t_{\text{e-He}}^{-1} - K(\mathrm{e},\mathrm{Kr},\overline{\mathrm{Kr}^*})n_\mathrm{e}\mathrm{Kr}E_\mathrm{Kr}^{\overline{\mathrm{Kr}^*}} \\
&\quad - K(\mathrm{e},\overline{\mathrm{Kr}^*},\mathrm{Kr}^+)n_\mathrm{e}\overline{\mathrm{Kr}^*}\left(\frac{3}{2}T_\mathrm{e} + E_{\overline{\mathrm{Kr}^*}}^{\mathrm{Kr}^+}\right) \\
&\quad - K(\mathrm{e},\mathrm{Kr},\mathrm{Kr}^+)n_\mathrm{e}\mathrm{Kr}\left(\frac{3}{2}T_\mathrm{e} + E_\mathrm{Kr}^{\mathrm{Kr}^+}\right) \\
&\quad - K(\mathrm{e},\mathrm{F}_2,\mathrm{F})n_\mathrm{e}\mathrm{F}_2\frac{3}{2}T_\mathrm{e} - K(\mathrm{e},\mathrm{Kr}_2^+,\mathrm{Kr})n_\mathrm{e}\mathrm{Kr}_2^+\frac{3}{2}T_\mathrm{e}
\end{aligned}
\tag{BA-16}
$$

with

$$
E_\mathrm{Kr}^{\overline{\mathrm{Kr}^*}} = \frac{K(\mathrm{e},\mathrm{Kr},\mathrm{Kr}^*)E_\mathrm{Kr}^{\mathrm{Kr}^*} + K(\mathrm{e},\mathrm{Kr},\mathrm{Kr}^{**})E_\mathrm{Kr}^{\mathrm{Kr}^{**}}}{K(\mathrm{e},\mathrm{Kr},\mathrm{Kr}^*) + K(\mathrm{e},\mathrm{Kr},\mathrm{Kr}^{**})}
$$

and

$$
K(\mathrm{e},\mathrm{Kr},\mathrm{Kr}^*) = K(\mathrm{e},\mathrm{Kr},\mathrm{Kr}^*) + K(\mathrm{e},\mathrm{Kr},\mathrm{Kr}^{**}). \qquad ???
$$

Since various molecules or atoms continue to react, their particle densities must be calculated separately.

B.2 Chemical reactions

As a result of the electrical discharge, a number of species are present in the "laser gas", which are briefly summarized as follows:

He	Kr	F_2	e
He^*	Kr^*	F	
He^+	Kr^{**}	F^-	
	Kr^+		

He^* and He^+ are, as the previous discussion has shown, present only in very low concentrations. The reason for this is the very high excitation and ionization potential. Therefore, helium will not be considered further in the following discussion.

Between the various krypton atoms and ions, as well as the F_2, F, and F^- particles, a number of reactions occur, the most important of which are the "laser reactions":

$$
\mathrm{Kr}^* + \mathrm{F}_2 \rightarrow \mathrm{KrF}^* + \mathrm{F} \tag{9}
$$

$$
\mathrm{Kr}^{**} + \mathrm{F}_2 \rightarrow \mathrm{KrF}^* + \mathrm{F} \tag{10}
$$

$$
\mathrm{Kr}^+ + \mathrm{F}^- + \mathrm{M} \rightarrow \mathrm{KrF}^* + \mathrm{M} \tag{11}
$$

The excited KrF^* has a radiative lifetime of approximately 11 ns (see below) and decays into Kr and F:

$$
\mathrm{KrF}^* \rightarrow \mathrm{Kr} + \mathrm{F} + h\nu. \tag{12}
$$

This process is stimulated in the laser arrangement.

In addition to these laser reactions, a series of other collision, emission, and absorption processes occur, which will also be discussed below. An important point in this context is that these additional processes not only influence the laser emission but also affect the discharge itself.

These processes can be divided into the following categories:

(a) Reactions between krypton species

(b) Laser reactions

(c) Deactivation reactions

(d) Trimer formation and deactivation

In Fig. 8, the potential diagram for Kr and the krypton compounds is schematically shown, from which the various reactions are visible.

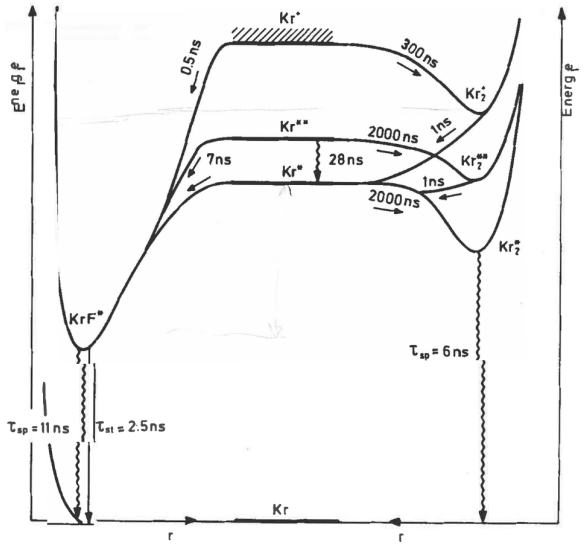


Figure 8: Potential diagram for the reactions of Kr in the laser gas

B.2.a Krypton–krypton reactions

First, we briefly discuss the krypton–krypton reactions that occur between the excited or ionized particles. In this context, reference is especially made to the comprehensive article by Werner et al. [15], which extensively discusses the formation of noble gas dimers:

$$\mathrm{Kr}^+ + 2\mathrm{Kr} \rightarrow \mathrm{Kr}_2^+ + \mathrm{Kr} \qquad K(\mathrm{Kr}^+, 2\mathrm{Kr}, \mathrm{Kr}_2^+) = 2.7 \times 10^{-31} \quad [15] \tag{13}$$

$$\mathrm{Kr}^{**} + 2\mathrm{Kr} \rightarrow \mathrm{Kr}_2^{**} + \mathrm{Kr} \qquad K(\mathrm{Kr}^{**}, 2\mathrm{Kr}, \mathrm{Kr}_2^{**}) = (2\text{–}5) \times 10^{-32} \quad [15] \tag{14}$$

$$\mathrm{Kr}^{*} + 2\mathrm{Kr} \rightarrow \mathrm{Kr}_2^{*} + \mathrm{Kr} \qquad K(\mathrm{Kr}^{*}, 2\mathrm{Kr}, \mathrm{Kr}_2^{*}) = (2\text{–}5) \times 10^{-32} \quad [15] \tag{15}$$

The characteristic time of reaction (13) is

$$\tau_{(13)} = \left(K(\mathrm{Kr}^+, 2\mathrm{Kr}, \mathrm{Kr}_2^+)\mathrm{Kr}^2\right)^{-1} \approx 300\,\mathrm{ns}$$

thus slow compared to the discharge time of about 40 ns. Reactions (14) and (15) can be treated together, since Kr_2^{**} decays very quickly (within ns) into Kr* and Kr, and the resulting Kr* eventually continues to react to form dimers as shown in reaction (15). The timescale for reactions (14)–(15) is therefore given by

$$\tau_{(14)\text{-}(15)} \approx \left(K(\mathrm{Kr}^*, 2\mathrm{Kr}, \mathrm{Kr}_2^*)\mathrm{Kr}^2\right)^{-1} \approx 2000\,\mathrm{ns}.$$

In Fig. 8, these times are indicated along the various reaction paths. All krypton–krypton reactions are thus slow compared to the excitation.

B.2.b Krypton–fluorine reactions

The inversion-generating process in the KrF laser is the reaction of excited or ionized krypton atoms with fluorine:

$$\mathrm{Kr^* + F_2 \rightarrow KrF^* + F;} \qquad K(\mathrm{Kr^*, F_2, KrF}) = 6.5 \times 10^{-10}\,\mathrm{cm^3/sec} \tag{16}$$

$$\mathrm{Kr^{**} + F_2 \rightarrow KrF^* + F;} \qquad K(\mathrm{Kr^{**}, F_2, KrF}) = ? = 6.5 \times 10^{-10}\,\mathrm{cm^3/sec} \tag{17}$$

$$\mathrm{Kr^+ + F^- + M \rightarrow KrF^* + M;} \qquad K = 2 \times 10^{-6}\,\mathrm{cm^3/sec} \tag{18}$$

The rate constant for reaction (16) was determined by Velarzco et al. [16] in a discharge afterglow setup. It is worth noting in this context that some authors, citing the same work [16], report quite different rate constants. The cited values range from $5.2 \times 10^{-10}\,\mathrm{cm^3/sec}$ to $10 \times 10^{-10}\,\mathrm{cm^3/sec}$.

As for the rate constants for reaction (17), no values are available in the literature. However, we assume that this reaction proceeds at least as quickly as reaction (16).

The reaction (18) is diffusion-limited, i.e., within the long-range Coulomb forces, there are enough neutral particles M such that the reaction partners $\mathrm{Kr^+}$ and $\mathrm{F^-}$ "diffuse toward each other". This results in the fact that with increasing external gas pressure, the reaction constant decreases. At the typical pressures of 2 bar He, Mangano et al. [16] reported a value of $2 \times 10^{-6}\,\mathrm{cm^3/sec}$.

In this context, the rates at which these reactions occur are important. Assuming a constant partial pressure of $\mathrm{F_2}$ (6 mbar), the half-life for reactions (16) and (17) is:

$$\tau_{(16)(17)} \approx 9\,\mathrm{ns}.$$

Even faster is the ion reaction (18). Here, typical times of 0.5 ns are expected for $\mathrm{F^-}$ particle densities of $10^{15}\,\mathrm{cm^{-3}}$, i.e., the formed Kr ions react immediately, while the excited krypton atoms take a little longer to react. During the excitation pulse of 40 ns, both reactions must, of course, be considered.

B.2.c Deactivation Reactions

The excitation reactions (16)–(18) compete with the deactivation processes, i.e., KrF* loses its excitation energy nonradiatively through energy transfer. The following processes are relevant here:

$$\mathrm{KrF^* + F_2 \rightarrow Kr + F + F} \qquad K(\mathrm{KrF^*, F_2, Kr}) = 5.7 \times 10^{-10}\,\mathrm{cm^3/sec} \quad [17] \tag{19}$$

$$\mathrm{KrF^* + Kr \rightarrow 2Kr + F} \qquad K(\mathrm{KrF^*, Kr, Kr}) = 10^{-12}\,\mathrm{cm^3/sec} \quad [17] \tag{20}$$

$$\mathrm{KrF^* + Kr + He \rightarrow 2Kr + F + He} \qquad K(\mathrm{KrF^*, KrHe, Kr}) = 5 \times 10^{-31}\,\mathrm{cm^6/sec} \quad [13, 18] \tag{21}$$

$$\mathrm{KrF^* + He + He \rightarrow Kr + F + 2He} \qquad K(\mathrm{KrF^*, HeHe, Kr}) = 5 \times 10^{-32}\,\mathrm{cm^6/sec} \tag{22}$$

$$\tau_{(19)} \approx 10\,\mathrm{ns}, \qquad \tau_{(20)} \approx 300\,\mathrm{ns}, \qquad \tau_{(21)(22)} \approx 7\,\mathrm{ns}.$$

The typical deactivation times listed above show that the lifetime is strongly limited by collisions.

$$\tau_{\mathrm{Deact}} = \left(\sum_i \frac{1}{\tau_i} \right)^{-1} \approx 2.6\,\mathrm{ns}$$

Without including radiative processes, the KrF* concentration will therefore closely follow the pump pulse. There are no literature values available for deactivation via electron collisions, so this process is not considered.

B.2.d Trimer reactions

The following two reactions are relevant:

$$\mathrm{KrF^* + 2Kr \rightarrow Kr_2F^* + Kr;} \qquad K(\mathrm{KrF^*, 2Kr, Kr_2F}) = 2.9 \times 10^{-31}\,\mathrm{cm^6/sec} \quad [17] \tag{23}$$

$$\mathrm{KrF^* + Kr + He \rightarrow Kr_2F^* + He;} \qquad K = ? \tag{24}$$

The half-life of reaction (23) is approximately 200 ns, so it can be neglected in comparison to the other faster processes. No literature values are available for reaction (24). If the corresponding rate constant were the same as for reaction (23), relatively short reaction times of only 12 ns would result. Since this is speculative, this process is not considered in our model.

The previous discussion has shown that the build-up rate of the inversion proceeds in comparable times to the collision deactivation. Two processes in deactivation are unclear: electron collision deactivation and trimer formation during Kr–He–KrF* collisions. Both processes could negatively affect the formation of KrF*.

B.3 Radiative processes

B.3.a Spectroscopic fundamentals

The potential levels of all noble gas halides are very similar and will be briefly explained below using KrF as an example (see Fig. 9). The ground state of noble gas halides is typically antibinding. The overlap of the ^1S state of the noble gas with the ^2P ground state of the halogen atom results in a stable bond only in the case of XeCl ($\Delta E = 255\,\mathrm{cm^{-1}}$) and XeF ($\Delta E = 1100\,\mathrm{cm^{-1}}$). Depending on the arrangement of the ^2P orbitals, the ground state is split into a $^2\Sigma$ state and a $^2\Pi$ state. Due to the lesser overlap of the orbitals, the $^2\Pi$ state lies higher than the $^2\Sigma$ state. The $^2\Sigma$ state is also denoted as X, and the $^2\Pi$ state is denoted as A. The first excited state is formed by ionic bonding. The ionized noble gas has ^2P orbitals, while the negative halide ion now has ^1S states (closed electron shell). Accordingly, the upper level is also split into $^2\Sigma$ and $^2\Pi$ states, of which the $^2\Pi$ state lies higher. Ionic bonds are typically characterized by broad, deep potential curves, which only reach a horizontal profile at large interatomic distances. This scheme is found in all noble gas halides. How far the splitting of the Π states extends depends, of course, on the spin–orbit coupling of the specific molecule.

For laser emission, it is important whether the ionic state lies higher or lower than the corresponding covalent states. The higher the ionization potential of the noble gas, the more the ionic potential curve shifts upwards. For example, in ArBr and KrI, the corresponding curves just touch, whereas in HeI, the ionic state lies entirely above the covalent curve. As a result of predissociation or lack of inversion, no laser emission has been observed in such systems, meaning that a possible inversion is linked to the condition that the ionic state is the lowest. Fundamental spectroscopic studies can be found in [19, 20, 21].

The fluorescence spectrum of the noble gas halides is structured similarly, reflecting the similar potential schemes of these molecules. In the following, this spectrum will be discussed in more detail using KrF as an example (see Fig. 10).

Emission from the B ($^2\Sigma_{1/2}$) state can occur both to the X ($^2\Sigma_{1/2}$) state and to the A ($^2\Pi_{1/2,3/2}$) state. Since the A states are higher, the $B \rightarrow A$ transition is shifted to longer wavelengths compared to the $B \rightarrow X$ transition. The steep nature of the A state potential leads to a broader emission bandwidth for the $B \rightarrow A$ transition, as compared to the $B \rightarrow X$ transition. The greater orbital overlap in the case of the B and X states also results in a larger transition probability between these two levels than for the corresponding $B \rightarrow A$ transition. In KrF, the B and C states nearly coincide, so the emission spectra from these states are almost indistinguishable (for other noble gas halides, this may not be the case).

The transition from the D state to the ground state $D \rightarrow X$ has also been observed in a number of noble gas halides. For KrF, this transition occurs at 220 nm, at 260 nm for XeF, and at 199 nm for KrCl. However, the intensity of the $D \rightarrow X$ transition is much weaker than that of the $B \rightarrow X$ transition, as in most cases the D state rapidly relaxes to the B state. More precise values for this relaxation process are not yet known.

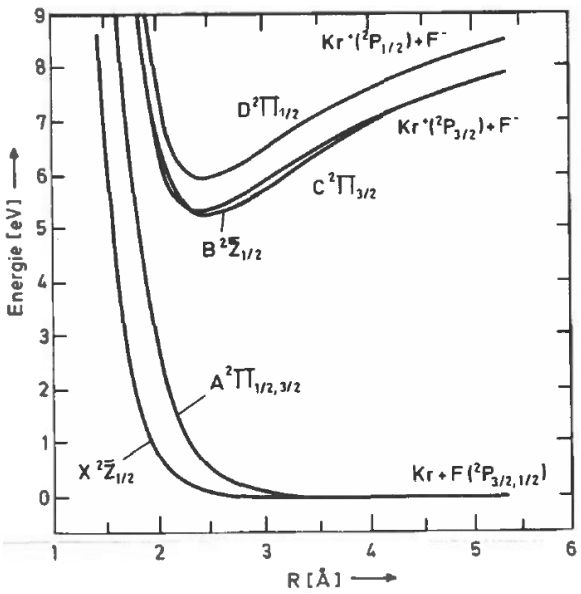


Figure 9: Calculated potential energy curves for the covalent and ionic states of KrF, taking into account spin–orbit interaction

The emission spectrum of the noble gas halides is quite complex and is significantly influenced by relaxation processes of the individual states. An additional complication arises because trimers form in the discharge gases, and their emission also falls within the spectral ranges being considered. For instance, in KrF, the $C \to A$, $D \to A$, and $B \to A$ emissions are overlapped by trimer emission, which is observed at approximately 400 nm in Kr_2F.

B.3.b Cross section for stimulated emission

One of the key laser parameters, the cross section for stimulated emission, is defined as:

$$\sigma(\nu) = \frac{A\lambda^2}{8\pi} g(\nu), \tag{B3-1}$$

where $A = \tau_{sp}^{-1}$ is the transition probability between the two levels (i.e., the Einstein coefficient A_{21}). The line shape factor $g(\nu)$ for Doppler-broadened lines has the form:

$$g_D(\nu) = \frac{2\ln 2}{\pi^{1/2}\Delta\nu_D} \exp\left(-\frac{4(\ln 2)(\nu-\nu_0)^2}{\Delta\nu_D^2}\right).$$

For $\nu = \nu_0$, the maximum cross section is given by $\sigma_D(\nu_0) = \frac{1}{4\pi}\left(\frac{\ln 2}{\pi}\right)^{1/2} \frac{\lambda^4}{c\Delta\lambda}$. For a Lorentzian profile (pressure broadening), the line shape is:

$$g_L(\nu) = \frac{\Delta\nu_L/2\pi}{(\nu-\nu_0)^2 + (\Delta\nu_L/2)^2};$$

and $\sigma_L(\nu_0) = \frac{A}{8\pi^2}\frac{\lambda^4}{c\Delta\lambda}$ is the the maximum cross section for a Lorentzian profile line.

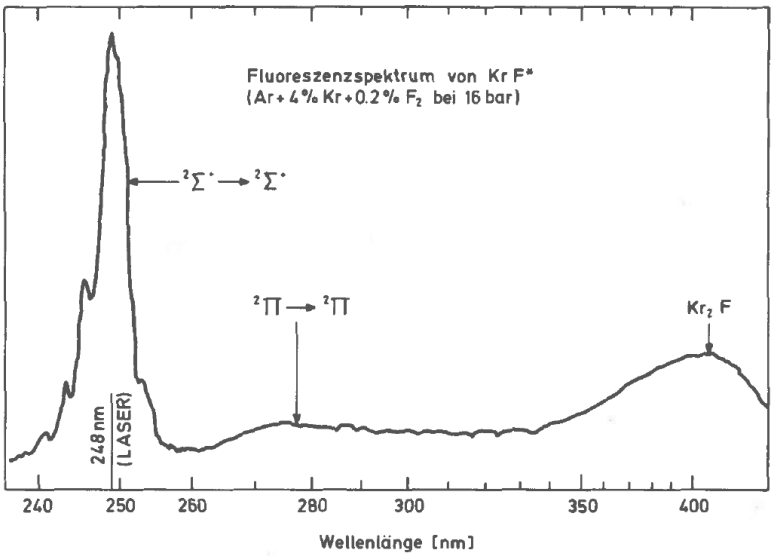


Figure 10: Fluorescence spectrum of KrF pumped by an electron beam [1].

The line shape for the KrF laser is determined – at least in the spectral regions of interest – not by pressure or temperature effects, but by the overlap integral of the B and X states. In Fig. 11a, the calculated potential curves with the corresponding vibronic states are shown according to [21]. The wave structure of the emission (see Fig. 11b) arises because the first vibronic states are also populated.

The cross section for bound–free transitions in the Franck–Condon region is calculated according to [22] by:

$$\sigma_{v'}(\nu) = \frac{8\pi^3 \nu}{3hc} \left|\mu_e(\bar{R})\right|^2 \left|\langle \psi_X^J | \psi_B^{v',J} \rangle\right|^2 \tag{B3-2}$$

where $\psi_B^{v,J}$ and ψ_X^J represent the radial wavefunctions of the B and X states, respectively, with rotational quantum number J. $\mu_e(R)$ is the dipole moment of the transition at the average internuclear distance $\bar{R}$. The lifetime τ for spontaneous emission is given by

$$A = \frac{1}{\tau} = \frac{64\pi^4}{3h} \left|\mu_e(\bar{R})\right|^2 \int \nu^3 \left|\langle \psi_X^J | \psi_B^{v',J} \rangle\right|^2 \mathrm{d}\nu. \tag{B3-3}$$

That is, it does not depend on ν, while σ is of course a function of the frequency. If different vibronic states contribute to the emission, then in the case of a Boltzmann occupation of these states, an average cross section is defined by:

$$\overline{\sigma(\nu)} = \frac{1}{q_{v'}} \sum_{v'} \sigma_{v'}(\nu) \exp\left(-\frac{E_{v'}}{kT}\right) \tag{B3-4}$$

where $q_{v'}$ is the degeneracy of the upper vibronic state v'.

The exact emission spectrum of KrF depends on the population of the different vibronic states and, thus, on the temperature of the laser gas. Nothing is known about the relaxation times of the vibronic states. However, it is assumed that these times are normally short in comparison to the laser pulse, so the line is homogeneously broadened.

In [21], the following value is calculated for a gas temperature of 360 K:

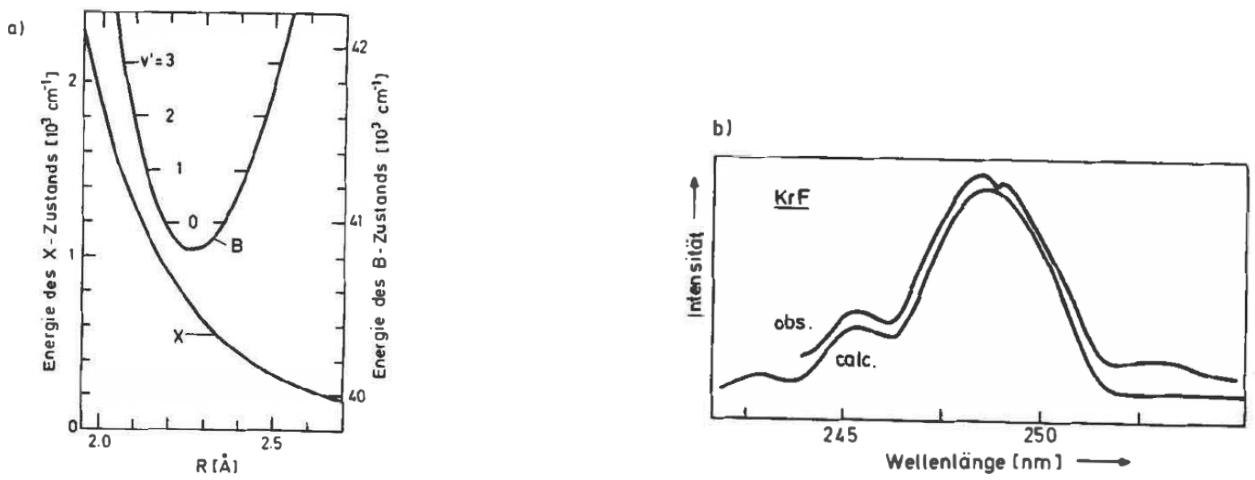


Figure 11: a) Calculated potential curves [21] for the $B \rightarrow X$ state. It should be noted that the zero point of the energy scales for the two states is different. b) Comparison of calculated vs. observed emission spectrum of the KrF laser transition

$$\sigma_{\max}\tau = 17 \pm 2 \quad (\text{Å}^2\,\text{ns})$$

Measuring the lifetime τ of KrF* is quite difficult due to the various fast-decaying quenching reactions. For example, Dunning and Hay [23] gave a value of 6.5 ns, while more recently, G.P. Quigley [17] measured a significantly larger value, which was also used in our calculations: $\tau = 11$ ns. With this, one finally obtains a cross section of $\sigma_{\max} = 1.6 \times 10^{-16}\,\text{cm}^2$.

In our model, this value for cross section is used, and it is further assumed that the line is homogeneously broadened.

B.3.c Absorption processes in the laser gas

The fluorescence or laser radiation of KrF is partially absorbed in the discharge gas, primarily by F_2, F^-, and Kr*/Kr**. The available laser energy is thereby significantly reduced, as will be shown in the further discussion.

F_2: From the work of Steunenberg and Vogel [24], an absorption cross section of $1.5 \times 10^{-20}\,\text{cm}^2$ can be derived. For typical laser gas mixtures containing 5 torr of F_2, an absorption of

$$\alpha = \left(1 - \frac{I}{I_0}\right) = 0.17$$

is calculated for a laser head length of 70 cm, meaning that 17% of the laser radiation is absorbed per pass through the laser head.

F^-: During the discharge, F^- is formed, which reacts very quickly to form KrF*. F^- has a very large absorption cross section [25]:

$$F^- + h\nu \rightarrow F + e \qquad \sigma = 5.6 \times 10^{-18}\,\text{cm}^2.$$

The exact concentration of F^- will be derived from the rate equations. The influence of F^- becomes clear when one considers that at densities of $10^{15}\,\text{cm}^{-3}$, 24% of the radiation is already absorbed over a length of 50 cm.

Kr*/Kr**: The absorption by Kr* and Kr** was calculated by H. Hyman [26]. The estimate yields:

$$\sigma_{Kr^*} = 1.3 \times 10^{-19}\ \mathrm{cm}^2$$
$$\sigma_{Kr^{**}} = 4.5 \times 10^{-18}\ \mathrm{cm}^2$$

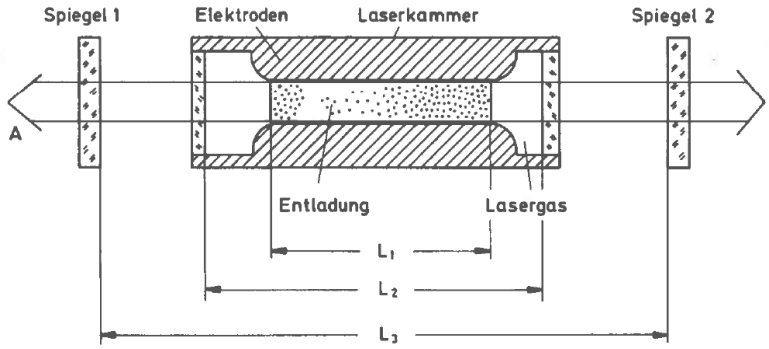
For Kr^{**}, these values are nearly the same as those for F^-. Again, it holds that even small densities of Kr^*/Kr^{**} lead to significant absorption. More accurate analyses can only be performed using detailed calculations.

B.3.d Laser equations

The rate equations model is used for the photon fluxes, as coherent effects in the KrF laser occur on a timescale of about 10^{-14} sec. A specific mode structure is not assumed. This is generally justified in most cases, as a multimode structure is typically observed in the KrF laser due to the high "gain" that occurs.

Fig. 12 schematically shows the cross section of the laser.

Before discussing the photon flux inside the resonator, the question of how the output power or energy outside the resonator relates to the photon flux inside the resonator will be addressed.



L_1 = length of the discharge
L_2 = length of the gas zone
L_3 = length of the resonator
A = cross-sectional area of the laser beam

Figure 12: Schematic setup of a KrF laser with external mirrors

(α) Output power as a function of the mirror parameters

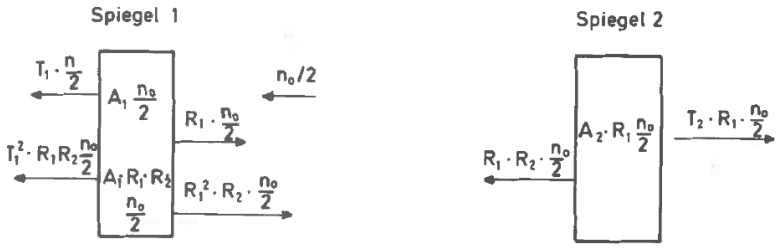
The measured energy at the output mirror is derived from the following considerations (see Fig. 13).

Assuming that at a time t_0, the photons (density n_0) are uniformly distributed over the resonator volume V, the transmission losses for the waves traveling to the left or to the right at mirror 1 of transmission T_1, absorption A_1, and reflection R_1 is:

$$\overleftarrow{N_1}(T_1) = \frac{N_0}{2}\left(T_1 + T_1(R_1R_2) + T_1(R_1R_2)^2 + \cdots\right) = \frac{N_0}{2}T_1\sum_{n=0}^{\frac{tc}{2L}}(R_1R_2)^n = T_1N_0\frac{1-(R_1R_2)^{ct/2L}}{2(1-R_1R_2)}.$$

$\frac{ct}{2L}$ is the number of passes.

Similarly, the following holds:



$A_{1/2}$ = absorption of mirror 1 / 2

$T_{1/2}$ = transmission of mirror 1 / 2

$R_{1/2}$ = reflection of mirror 1 / 2

$$A_{1/2} + T_{1/2} + R_{1/2} = 1$$

Figure 13: Coupling ratio in a passive resonator

$$\overrightarrow{N_1}(T_1) = R_2 T_1 N_0 \frac{1-(R_1R_2)^{ct/2L}}{(1-R_1R_2)2},$$

and the total transmission losses up to time t from T_1 are:

$$N(T_1) = \overleftarrow{N}(T_1) + \overrightarrow{N}(T_1) = (1+R_2)T_1N_0\frac{1-(R_1R_2)^{ct/2L}}{2(1-R_1R_2)}. \tag{B3-5}$$

Correspondingly, for the transmission losses at T_2:

$$N(T_2) = (1+R_1)T_2N_0\frac{1-(R_1R_2)^{ct/2L}}{2(1-R_1R_2)}.$$

The absorption losses result from similar considerations:

$$N(A_{\underset{2}{1}}) = A_{\underset{2}{1}}(1+R_{\underset{1}{2}})N_0\frac{1-(R_1R_2)^{ct/2L}}{2(1-R_1R_2)}. \tag{B3-6}$$

Within the resonator, the photon density decreases:

$$N_{\mathrm{i}} = N_0(R_1R_2)^{\frac{tc}{2L}} = N_0\exp\left[\frac{tc}{2L}\ln(R_1R_2)\right]. \tag{B3-7}$$

As one can easily verify, it must, of course, hold:

$$N(T_1) + N(T_2) + N(A_1) + N(A_2) + N_{\mathrm{i}} = N_0.$$

The previous discussion assumed a passive resonator, where at time t_0, there were N_0 photons in the resonator. However, due to the stimulated processes in the laser, N_0 is constantly changing. The transition to the "active" resonator can be made by the following consideration: the change in the photons coupled out at mirror T_1 depends only on the instantaneous particle density $N_{\mathrm{i}}(t)$ in the resonator:

$$\frac{\mathrm{d}N(T_1)}{\mathrm{d}t} = -\frac{T_1}{1-R_1R_2}\frac{1+R_2}{2}\frac{c}{2L}\ln(R_1R_2)N_\mathrm{i}(t). \tag{B3-8}$$

This equation now holds generally, as it doesn't matter how we arrive at $N_\mathrm{i}(t)$, i.e., in the general case, the energy coupled out up to time t can be found by integration:

$$N(T_1)(t) = \frac{(1+R_2)T_1c|\ln R_1R_2|}{4L(1-R_1R_2)}\int_0^t N_\mathrm{i}\,\mathrm{d}t, \tag{B3-9}$$

from which (B3-5) results, if N_i is substituted from (B3-7). Now, proceeding from the particle numbers to the particle densities:

$$n = \frac{N}{L_3A} \qquad (L_3 = \text{resonator length}, A = \text{laser cross section})$$

the output power is given by:

$$P = \frac{T_1(1+R_2)c|\ln R_1R_2|h\nu A}{4(1-R_1R_2)}n_\mathrm{i}, \tag{B3-10}$$

and the output energy by:

$$E = \int P\,\mathrm{d}t. \tag{B3-11}$$

The transition from the passive to the active resonator, as presented here, can only be made if the changes in photon number N occur slowly compared to the resonator's dwell time. It is assumed that the photon density is the same throughout the entire resonator. However, this can only be considered an approximation in the case of the KrF laser. For an exact calculation, the waves traveling to the left and right must be treated separately.

(β) Photon flux in the resonator

The amplification after m laser passes is: $V(mL_3) = \exp\left(\sigma\Delta N\frac{L_1}{L_3}mL_3\right)$, since the amplification is along the partial segment L_1. For m passes, the time $\Delta t = \frac{mL_3}{c}$ is needed, and the amplification over time Δt is $V(\Delta t) = \exp\left(\sigma\Delta N\frac{L_1}{L_3}c\Delta t\right)$.

Then:

$$\frac{\mathrm{d}n}{\mathrm{d}t} = \lim_{\Delta t\to 0}\frac{n(t+\Delta t)-n(t)}{\Delta t} = \lim\frac{n(t)V(\Delta t)-n(t)}{\Delta t} = \sigma c\frac{L_1}{L_3}\Delta N n$$

where ΔN represents the inversion density in the laser.

The photon number losses due to output mirrors are calculated from (B3-7) as:

$$\left.\frac{\mathrm{d}n}{\mathrm{d}t}\right|_{\text{mirror loss}} = -\frac{c}{2L}|\ln(R_1R_2)|n.$$

Finally, the absorption processes in the laser gas must also be considered (see B3.c). An analogous treatment as for amplification leads to:

$$\left.\frac{\mathrm{d}n}{\mathrm{d}t}\right|_{\text{absorption}} = -\left(\sigma_{\mathrm{F_2}}\mathrm{F_2}\frac{L_2}{L_3} + \sigma_{\mathrm{F}^-}\mathrm{F}^-\frac{L_1}{L_3} + \sigma_{\overline{\mathrm{Kr}^*}}\overline{\mathrm{Kr}^*}\frac{L_1}{L_3}\right)n,$$

where $\sigma_{\overline{\mathrm{Kr}^*}}$ is defined by: (see Chapter B1/d)

$$\sigma_{\overline{\mathrm{Kr}^*}} = \sigma_{\mathrm{Kr}^{**}}\frac{a(T_\mathrm{e})}{1+a(T_\mathrm{e})}.$$

Including the spontaneous emission (see, for example, [26]), the photon density $n = \frac{N}{FL_3}$ is given by:

$$\frac{\mathrm{d}n}{\mathrm{d}t} = \sigma_{\mathrm{KrF}} c \mathrm{KrF} \frac{L_1}{L_3}(n+1) - \frac{c}{2L_3}|\ln(R_1 R_2)|n - cn\left[\sigma_{\mathrm{F_2}}\mathrm{F_2}\frac{L_2}{L_3} + \sigma_{\mathrm{F^-}}\mathrm{F^-}\frac{L_1}{L_3} + \sigma_{\overline{\mathrm{Kr^*}}}\overline{\mathrm{Kr^*}}\frac{L_1}{L_3}\right], \tag{B3-12}$$

from which the laser power and laser energy can be calculated according to (B3-10) and (B3-11).

C Parametric investigation

C.1 System of equations

Table 1 summarizes all the processes that, based on the previous discussion, influence the emission behavior of the KrF laser. Also included are the reactions that have only a very weak influence, such as the dimer formation of Kr. Kr* and Kr** are treated separately so that small deviations from thermal equilibrium for these two species can be accounted for. As a result, there are changes to the previous nomenclature, which are necessary for better handling of the program. The labels listed here for the individual quantities are the same as those used in the program.

Table 1 summarizes the following processes:

(a) Excitation and ionization of Kr, He, and F_2 in the discharge (Reactions 1–10).

(b) Reactions of metastable atoms and ions in the presence of F_2 and F^-, leading to KrF* formation (Reactions 25–27).

(c) Reactions involving dimers (Reactions 11, 15–16).

(d) Deactivation of KrF* (Reactions 32–36).

The processes under (a) and for Reaction 11 depend on the electron energy E_e, where we assume that it follows a Maxwell distribution. The effective electron temperature is $T_e = \frac{2}{3}E_e k^{-1}$.

The relaxation of T_e through e–He collisions is phenomenologically described by Reaction 19. The other processes listed in Table 1 are:

(e) Stimulated and spontaneous emission of KrF* (Reactions 30–31).

(f) Absorption of photons by F_2, F^-, and Kr_2^+ (Reactions 37–42).

(g) Losses due to mirrors and potential other processes, such as scattering within the resonator (Reactions 44–46).

The complete system of equations for the laser consists of two equations for the electrons (particle number and temperature) B1-15 and B1-16, equations for the photons B3-12, as well as several equations for the different particle types. In Table 2, the equations are listed as they actually appear in the program. Changes in the notation could not be avoided, but they are indicated for a better understanding of the program.

Table 1:

Column 1 describes the reaction index in the program. Column 2 specifies which reaction is involved. Under the "Rate Constant" section, it lists how this rate constant is calculated in the specific case. The "Parameters" column lists constants primarily derived from literature studies. The last column shows the correlation coefficient, which indicates by what percentage the result changes if the parameter in Column 4 is increased by + 100%.

The indices a and b mean:

(a) This value is obtained by averaging over a Maxwell distribution for the electrons. The cross section σ is given by: $\sigma(E) = 0$ for $E \leq E_0$; $\sigma(E) = \alpha(E - E_0)$ for $E > E_0$.

(b) E_0 is determined through other steps.

(c) L is the optical resonator length ($L = L_3$ in B3.d).

n	Reaction	Rate Constant K_n	Parameters	Correlation Coefficients
1	$Kr + e \rightarrow Kr^* + e$	$K = f(E_0, \alpha = \frac{d\sigma}{dE})$[a]	$E_0 = 9.95\,eV \quad \alpha = 11 \times 10^{-18}\,\frac{cm^2}{eV}$	$r_{E_0} = -152 \quad r_\alpha = 7$
2	$Kr + e \rightarrow Kr^{**} + e$	$K = f(E_0, \alpha = \frac{d\sigma}{dE})$[a]	$E_0 = 10.6\,eV \quad \alpha = 2 \times 10^{-17}\,\frac{cm^2}{eV}$	$r_{E_0} = 28 \quad r_\alpha = 10$
3	$Kr^* + e \rightarrow Kr^{**} + e$	$K = f(E_0, \alpha = \frac{d\sigma}{dE})$[a]	$E_0 = 0.65\,eV \quad \alpha = 1.5 \times 10^{-15}\,\frac{cm^2}{eV}$	$r_{E_0} = (-)$[b] $\quad r_\alpha = 0$
4	$He + e \rightarrow He^* + e$	$K = f(E_0, \alpha = \frac{d\sigma}{dE})$[a]	$E_0 = 19.8\,eV \quad \alpha = 10^{-18}\,\frac{cm^2}{eV}$	$r_{E_0} = 0 \quad r_\alpha = 0$
5	$Kr + e \rightarrow Kr^+ + 2e$	$K = f(E_0, \alpha = \frac{d\sigma}{dE})$[a]	$E_0 = 14\,eV \quad \alpha = 2.6 \times 10^{-17}\,\frac{cm^2}{eV}$	$r_{E_0} = 28 \quad r_\alpha = -5$
6	$Kr^* + e \rightarrow Kr^+ + 2e$	$K = f(E_0, \alpha = \frac{d\sigma}{dE})$[a]	$E_0 = 4.05\,eV \quad \alpha = 2 \times 10^{-16}\,\frac{cm^2}{eV}$	$r_{E_0} = (-)$[b] $\quad r_\alpha = -2$
7	$Kr^{**} + e \rightarrow Kr^+ + 2e$	$K = f(E_0, \alpha = \frac{d\sigma}{dE})$[a]	$E_0 = 3.4\,eV$	$r_{E_0} = (-)$[b] $\quad r_\alpha = -2$
8	$Kr^{**} + e \rightarrow Kr^* + e$	$K_3 \exp(0.65/kT_e)$	see rate 3	
10	$F_2 + e \rightarrow F + F^-$	$K_0 \exp(-E_0/kT_e)$	$E_0 = 0.41\,eV \quad K_0 = 3 \times 10^{-9}\,\frac{cm^3}{sec}$	$r_{E_0} = -2 \quad r_{K_0} = 10$
11	$Kr_2^+ + e \rightarrow Kr^{**} + Kr$	$K_0 \exp\left(-\frac{1}{2}\ln T_e\right)$	$K_0 = 2.4 \times 10^{-7}\,cm^3 eV^{1/2} sec^{-1}$	$r_{K_0} = 0$
15	$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	K	$K = 2.7 \times 10^{-31}\,cm^6/sec$	$r_K = 0$
16	$Kr^* + 2Kr \rightarrow Kr_2{}^* + Kr$	K	$K = 4.5 \times 10^{-32}\,cm^6/sec$	$r_K = 0$
18	$Kr^{**} \rightarrow Kr^* + h\nu$	$K = 1/\tau_{Kr^{**}}$	$\tau_{Kr^{**}} = 2.76 \times 10^{-8}\,sec$	$r = 0$
19	$He + e_S \rightarrow He + e_L$	$K = 1/\tau_{e\text{-}He}$	$\tau_{e\text{-}He} = 1.5 \times 10^{-9}\,sec$	$r = 28$
25	$Kr^* + F_2 \rightarrow KrF^* + F$	K	$K = 6.5 \times 10^{-10}\,cm^3/sec$	$T_K = 14$
26	$Kr^+ + F^- \rightarrow KrF^*$	equiv. 2 particle rate K	$K = 2 \times 10^{-6}\,cm^3/sec$	$T_K = 0$
27	$Kr^{**} + F_2 \rightarrow KrF^* + F$	K	$K = 6.5 \times 10^{-10}\,cm^3/sec$	$r_K = 7$
30	$KrF^* + nh\nu \rightarrow Kr + F + (n+1)h\nu$	stim. emission K	$K = 4.6 \times 10^{-6}\,cm^3/sec$	$r_K = 28$
31	$KrF^* \rightarrow Kr + F + h\nu$	spont. emission $K = 1/\tau_{KrF^*}$	$\tau_{KrF^*} = 1.1 \times 10^{-8}\,sec$	$r = 14$
32	$KrF^* + F_2$ (deactivation)	K	$K = 5.7 \times 10^{-10}\,cm^3/sec$	$r_K = -7$
33	$KrF^* + Kr$ (deactivation)	K	$K = 10^{-12}\,cm^3/sec$	$r_K = 0$
34	$KrF^* + 2Kr$ (deactivation)	K	$K = 6.5 \times 10^{-32}\,cm^6/sec$	$r_K = -3$
35	$KrF^* + Kr + He$ (deactivation)	K	$K = 5 \times 10^{-31}\,cm^6/sec$	$r_K = -7$
36	$KrF^* + 2He$ (deactivation)	K	$K = 5 \times 10^{-32}\,cm^6/sec$	$r_K = -3$
37	$Kr^{**} + h\nu \rightarrow Kr^+ + e$	K	$K = 1.35 \times 10^{-7}\,cm^3/sec$	$r_K = -21$
38	$Kr^* + h\nu \rightarrow Kr^+ + e$	K	$K = 3.9 \times 10^{-9}\,cm^3/sec$	$r_K = 0$
40	$F_2 + h\nu$ (absorption)	$K\,(= \sigma_{abs} c)$	$K = 4.5 \times 10^{-10}\,cm^3/sec$	$r = -7$
41	$F^- + h\nu \rightarrow F + e$	$K\,(= \sigma_{abs} c)$	$K = 1.68 \times 10^{-7}\,cm^3/sec$	$r = -3$
42	$Kr_2^+ + h\nu \rightarrow Kr^+ + Kr$	$K\,(= \sigma_{abs} c)$	$K = 4.8 \times 10^{-8}\,cm^3/sec$	$r = 0$
44	reflection losses[c]	K	$K = -c\ln(R_1 R_2)/L$	$R_1 = 0.1 \quad R_2 = 0.99$
45	distributed losses	K	$K = -c\ln(1 - DL)/L$	$L = 75\,cm \quad DL = 0.0$
46	transmission	K	$K = -\frac{cT_1(1+R_2)\ln(R_1 R_2)}{4(1-R_1 R_2)L}$	

Table 2: system of equations for the KrF laser

1. $\dot{n}_e = r_5 + r_6 + r_7 - r_{10} - r_{11} + r_{37} + r_{38} + r_{41}$
2. $$\begin{aligned} k\dot{T}_e &= \frac{2}{3}\dot{E}_e/n_e - \dot{n}_e T_e/n_e \\ &= P + \frac{2}{3n_e}\big[-r_1 E(\mathrm{Kr}, \mathrm{Kr}^*) - r_2 E(\mathrm{Kr}, \mathrm{Kr}^{**}) - r_3 E(\mathrm{Kr}^*, \mathrm{Kr}^{**}) - r_4 E(\mathrm{He}, \mathrm{He}^*) \\ &\quad - r_5(E(\mathrm{Kr}, \mathrm{Kr}^+) + T_e) - r_6(E(\mathrm{Kr}^*, \mathrm{Kr}^+) + T_e) - r_7(E(\mathrm{Kr}^{**}, \mathrm{Kr}^+) + T_e) \\ &\quad + r_8 E(\mathrm{Kr}^*, \mathrm{Kr}^{**}) + r_{37}(E(h\nu) - E(\mathrm{Kr}^{**}, \mathrm{Kr}^+) - T_e) \\ &\quad + r_{38}(E(h\nu) - E(\mathrm{Kr}^*, \mathrm{Kr}^+) - T_e) + r_{41}(E(h\nu) - EA(\mathrm{F}^-) - T_e)\big] - \dot{n}_e T_e/n_e \end{aligned}$$
3. $\dot{\mathrm{Kr}^+} = r_5 + r_6 + r_7 - r_{15} - r_{26} + r_{37} + r_{38}$
4. $\dot{\mathrm{He}^*} = r_4$
5. $\dot{\mathrm{Kr}^*} = r_1 - r_3 - r_6 + r_8 - r_{16} + r_{18} - r_{25}$
6. $P = I(t)V(t)$
7. $\dot{\mathrm{KrF}^*} = r_{26}$ — KrF* molecules formed by ion recombination
8. $\dot{\mathrm{Kr}_2^*} = r_{16}$
9. $\dot{\mathrm{KrF}^*} = r_{31}$ — loss of KrF* through spontaneous emission
10. $\dot{\mathrm{Kr}^{**}} = r_2 + r_3 - r_7 - r_8 + r_{11} - r_{18} - r_{27} - r_{37}$
11. $\dot{\mathrm{KrF}^*} = r_{25} + r_{26} + r_{27} - r_{30} - r_{31} - r_{32} - r_{33} - r_{34} - r_{35} - r_{36}$
12. $\dot{n_\phi} = r_{46}$ — photons lost through outcoupling
13. $\dot{n_\phi} = r_{37} + r_{38}$ — photons lost through photoionization
14. $\dot{n_\phi} = \frac{L_1}{L_3}(r_{30} - r_{37} - r_{38} - r_{41} - r_{42}) - \frac{L_2}{L_3} r_{40} - r_{44} - r_{45}$
15. $\dot{\mathrm{KrF}^*} = r_{25} + r_{27}$ — KrF* molecules formed by Kr* and Kr** reactions
16. $\dot{\mathrm{KrF}^*} = r_{32} + r_{33} + r_{34} + r_{35} + r_{36}$ — loss of KrF* due to deactivation
17. $\dot{n_\phi} = [L_2 r_{40} + L_1(r_{41} + r_{42})]/L_3$ — photons lost by absorption in F_2, F^-, Kr_2^+ reactions
18. $\dot{\mathrm{F}^-} = r_{10} - r_{26} - r_{41}$
19. $\dot{\mathrm{Kr}_2^+} = r_{15} - r_{11}$
20. $\dot{\mathrm{KrF}^*} = r_{27}$ — KrF* molecules formed by Kr** reactions

Table 2 contains 20 particle types, whose time dependence can be tracked.

The integration of this system of coupled first-order nonlinear differential equations is performed using the IMSL subroutine GEAR.

C.2 Results

For the calculations, experimental parameters from the "Mephisto" laser system were used as the basis. "Mephisto" is a UV-preionized TEA (Transversely Excited Atmospheric) laser, operated with a fast Blumlein discharge. The discharge volume is 150 cm^3 (width 1 cm, height 3 cm, length 50 cm), and the length of the chamber is 75 cm. The resonator length can be reduced to 75 cm. The maximum total pressure of the laser gas in the chamber is 2.5 bar.

The current and voltage across the discharge follow the profile shown in Fig. 14, from which the pump power $P(t)$ is easily calculated. $P(t)$ is shown as a dashed line in Fig. 14.

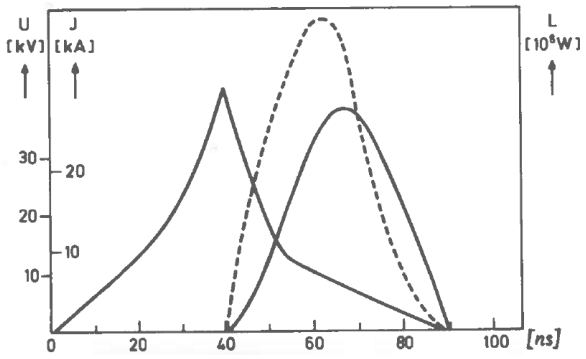


Figure 14: Current and voltage curve of the discharge in the laser system "Mephisto". Electron distance 3 cm, gas pressure 1.5 bar ($90 : 9.5 : 0.5 = \mathrm{He} : \mathrm{Kr} : \mathrm{F_2}$).

The main discharge lasts about 50 ns, with peak currents exceeding 26 kA. Using our program, we have calculated a total energy of about 750 mJ under these conditions, which corresponds to an efficiency of over 14%. However, this efficiency refers only to the energy coupled in during the first 50 ns. The actual observed energy and efficiency are significantly lower. A maximum of 300 mJ is measured, which corresponds to an efficiency of ~ 5.5%. The following will investigate which parameters account for these differences.

C.2.a Dependence of the laser yield η on the accuracy of the rate constant

A measure of the dependence of the yield on rate constants is the correlation coefficient r:

$$r = \left(\frac{\Delta\eta}{\eta} \Big/ \frac{\Delta K}{K} \right) \cdot 100\%$$

It indicates by how much the laser yield changes when the rate constant K is changed by ΔK. In Table 1, the correlation coefficients associated with the rate constants are shown in parentheses. Initially, it is evident that of the 33 parameters, 11 play virtually no role ($r \leq 1\%$). About 14 parameters have a relatively weak influence ($2\% \leq r \leq 10\%$). Only if all these parameters had the same systematic error – which is not the case – would their influence be significant. The remaining 8 parameters have a more or less strong effect on the results:

- 2 parameters with $r = 14\%$
- 5 parameters with $21\% \leq r \leq 28\%$
- 1 parameter with $r = 152\%$

and these parameters need to be discussed in more detail.

The influence of the excitation energies is generally very strong. However, since they are known with a precision of a few percent from electron beam measurements, deviations in the calculations cannot be attributed to these. This leaves 5 parameters that significantly influence the laser emission. These are Reaction 15 (KrF^* formation from Kr^* and F_2: 14%), stimulated emission with 28%, spontaneous lifetime with 14% (Reactions 30, 31), absorption of Kr^{**} (Reaction 37), and electron thermalization time (Reaction 19). Finally, Reaction 10 (F_2-electron attachment) must also be considered, as its reaction rate was only estimated (see Chapter B1.c).

Among these 6 parameters, 5 are relatively well known (Reactions 15, 10, 30, 31, 37), which in this context means an uncertainty of ±30%, i.e., they will influence the result with an uncertainty of no more than 40%. The most critical point is the e–He thermalization with $r = 28\%$, as this clearly demonstrates the strong influence that the electron

distribution has on the result. In our model, a Maxwell distribution of the electrons is assumed, which can only be an approximation. The actual electron distribution, as must be calculated from the Boltzmann equation, contains fewer fast electrons than the corresponding Maxwell distribution. However, it is possible to try to minimize the impact of this incorrect assumption by adjusting the relaxation rates.

Furthermore, a more detailed analysis shows that the correlation coefficient for the thermalization time $\tau_{\text{e-He}}$ strongly depends on $\tau_{\text{e-He}}$ itself. For example, if $\tau_{\text{e-He}}$ is reduced to 1 ns, r_τ becomes quite significant, reaching 50%.

Another unknown factor is the distributed losses within the resonator. Experiments with different resonator quality factors should provide insight into this.

The previous discussion has shown that only a few parameters contribute significantly to larger effects. Most parameters are either very well known, or their uncertainties are negligible. The latter case occurs because the corresponding processes are either very fast or very slow relative to the discharge duration. An example of the former is KrF* formation and deactivation reactions, while the latter applies to processes like dimer formation.

C.2.b Parametric investigation

In our model, the thermalization time $\tau_{\text{e-He}}$ was used as an adjustment parameter because the description of the electron distribution is only an approximation. By decreasing the $\tau_{\text{e-He}}$ value, the electron gas is cooled more effectively, and the number of fast electrons is reduced. This adjustment parameter had to be reduced to 0.3 ns to align the observed energy (and efficiency) with the calculated values. Additionally, the losses within the resonator were increased to 20% per pass (as discussed below).

Figs. 15a–c show typical results of the program. In Fig. 15a, the concentration of species contributing to KrF* (Kr*, Kr**, Kr^+) is plotted. The increase in Kr^+ at around 30 ns can be attributed to the photoionization of Kr** in the laser field, since the laser begins oscillating at around 25 ns (see Fig. 15c). At the moment the oscillation starts, the KrF* concentration sharply decreases and then stabilizes at a roughly constant value. With this adjusted parameter, further calculations were performed.

Specifically, it was investigated how the laser energy and efficiency depend on various parameters such as the input electrical energy, gas composition, and mirror reflectivities. The results are summarized in Figs. 16a–c, which will be discussed below. In Fig. 16a, the efficiency is plotted as a function of electrical energy. The efficiency approaches a constant value of about 6% at high pump energies. This maximum value is influenced by the various assumptions in our model and fluctuates between 5% and 9%. However, the general trend of the curve remains the same. After reaching the threshold at about 2.0 joules of pump energy, the efficiency rises steeply and levels off at about four times the threshold pump energy P_s. Notably, P_s is relatively high, which reflects the fact that deactivation processes play a significant role in this laser. The electrical threshold pump power per laser cross-sectional area is $\sim 16\,\text{MW/cm}^2$ (excitation duration ~ 40 ns).

In Fig. 16b, the dependence of the laser efficiency on the fluorine partial pressure is shown. Maximum values are reached at around 0.8% F_2 partial pressure, and the maximum is relatively broad. This trend, as well as the optimal values, are actually observed in experiments. The steep increase in efficiency with increasing F_2 partial pressure can be qualitatively explained by the faster formation of KrF under these conditions. The decrease in energy, on the other hand, is attributed to the stronger photon absorption by F_2.

As a final example, the laser output was calculated as a function of the mirror reflectivities (Fig. 16c). If the reflectivity of the end mirror is assumed to be 100%, the laser energy reaches its maximum at around 10–20% reflectivity of the coupling mirror (curve ∘). For higher feedback levels, the laser energy decreases again. This behavior is due to the fact that photons are more strongly absorbed at longer resonator dwell times. The experimentally observed curve, shown as a dashed line in Fig. 16c, follows this trend. The cause of the discrepancy at higher reflectivities is likely that stronger losses occur in the resonator than were assumed in the calculation. The scattering and refraction losses discussed in the previous section may be responsible for this. If these losses are assumed to be 20% per laser pass, the curve marked with ⊗ in Fig. 16c is obtained. It indeed represents the observed behavior quite well. Therefore, in

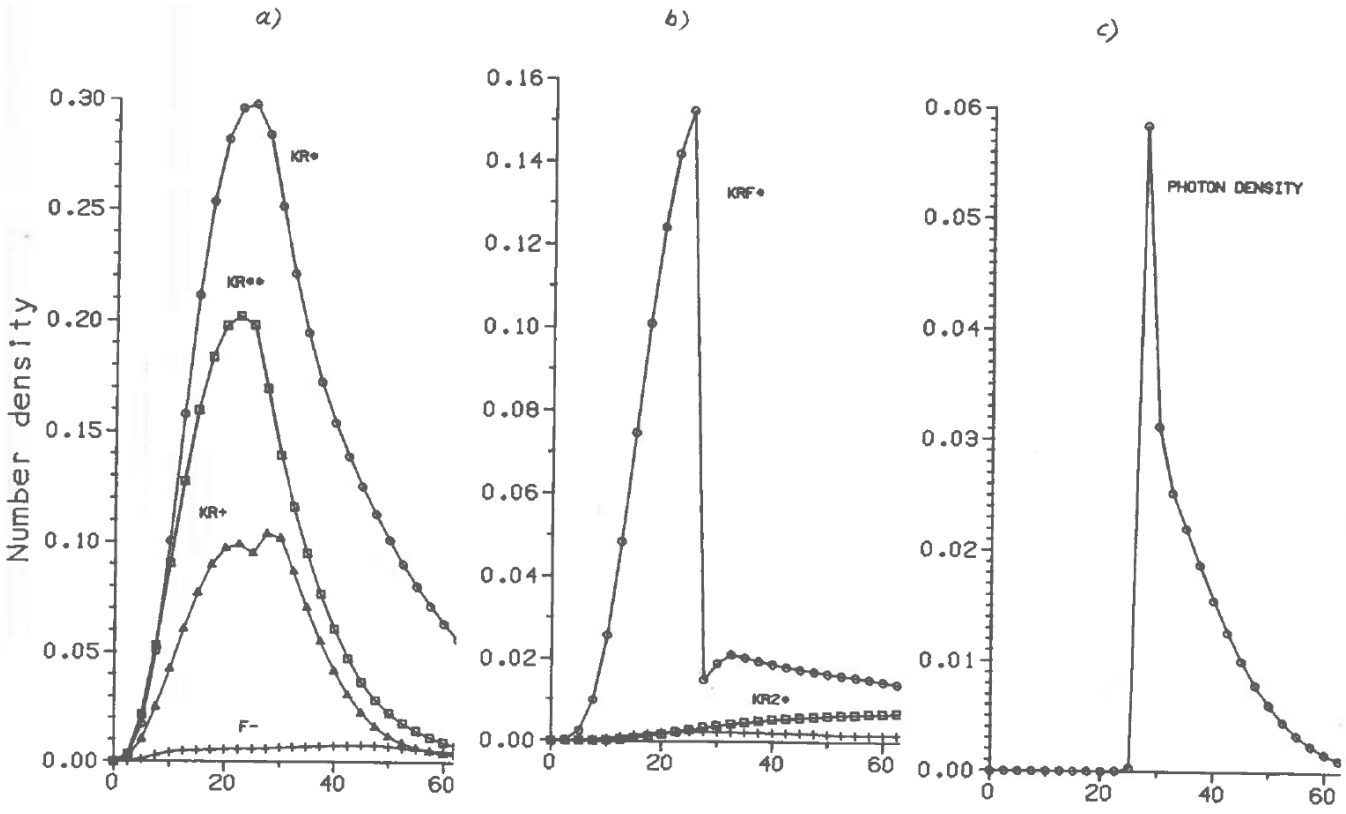


Figure 15: Temporal behavior during the discharge (particle densities in 10^{16} cm^{-3}, time in ns). For these calculations, the parameters from Table 1, $\tau_{\text{e-He}} = 0.3$ ns, $r_{40} = 4.6 \times 10^{-6}$ cm^3/sec and DL (distributed losses) = 0.2 were set.

these lasers, we will have to account for higher losses during passage through the laser medium than were previously assumed.

C.2.c Reaction flows

Finally, the reaction flows of the individual particle types are illustrated. They are shown in Fig. 17. The given values represent particle densities in units of 10^{16} cm^{-3}. A surprising result is the high proportion of the ionic reaction in the formation of KrF*. Nearly 30% are formed in this way. Another interesting finding is that only about 30% of the KrF* molecules contribute to the laser energy. The remaining energy is either spontaneously emitted (21%), converted into heat (31%), or reabsorbed (7%).

D Summary

In this work, we initially discussed the most important reaction steps in the KrF laser. For calculating the rate constants of electron collisions, it was assumed as a first approximation that the electron distribution can be described by a temperature T_e, i.e., it follows a Maxwell distribution. The energy absorbed by the laser gas was taken from the experiment (current and voltage measurements). The discussion of the reaction kinetics revealed that the formation of Kr species dimers plays no significant role. The most important processes – KrF* formation and deactivation – occur at nearly the same rate. The lifetime of KrF* molecules in typical gas mixtures is less than 2 ns (without laser radiation). The radiation field is described using the classical balance equations. The laser transition is considered as homogeneously broadened. Inhomogeneities in the excitation density across the laser cross section are not considered

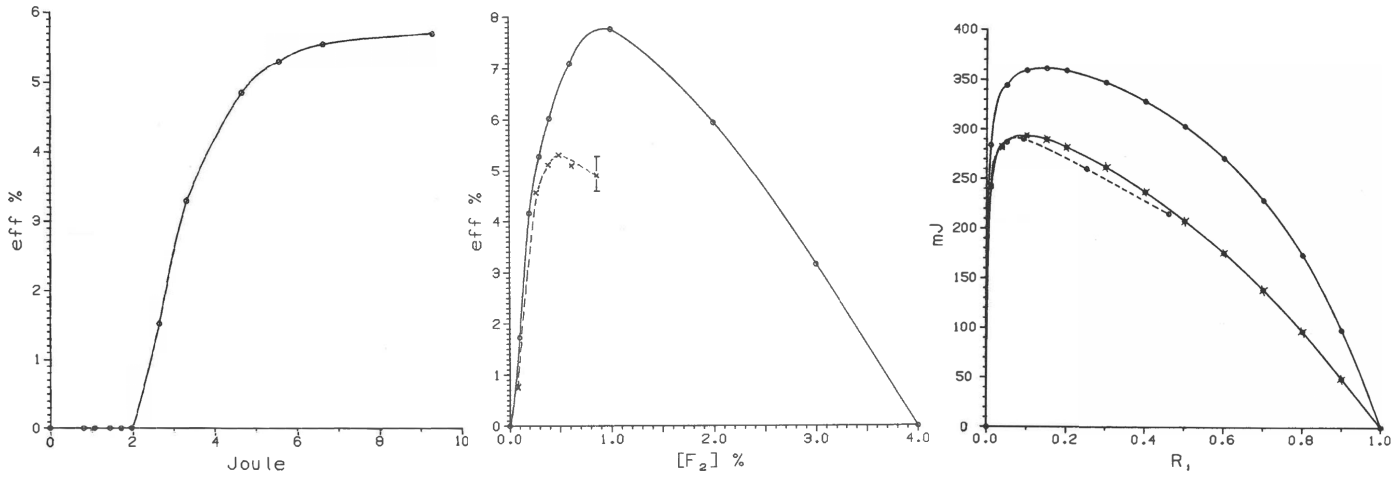


Figure 16: a) Yield of the KrF laser, based on the coupled electrical energy (for parameters, see text). b) Calculated output energy as a function of the fluorine partial pressure compared to experimentally found values (×). c) Calculated and experimentally determined laser energy as a function of the reflectivity of the output mirror.

in this model.

A parametric study showed that of the 31 different parameters that are included in the model, only four critically affect the results. These are:

(a) electron attachment to F_2 molecules

(b) electron temperature or electron distribution

(c) spontaneous lifetime or cross section for stimulated emission, and

(d) losses within the resonator.

The greatest uncertainty in the model lies in the choice of the correct electron distribution. If the normal e–He thermalization time is assumed, the yield is found to be 2.5 times too high. As an adjustment parameter, this thermalization time was chosen. When it is reduced by a factor of 5, the observed maximum energy values are obtained. The dependence of output energy on resonator quality can only be described correctly if additional losses in the resonator are assumed (20% per pass). These losses may be caused by pressure waves or by the lens properties of the plasma.

An improvement of the model can be achieved by calculating the actual electron distribution using the Boltzmann equation. However, the systematic errors have almost no impact on relationships that describe the laser energy as a function of pump energy, gas composition, and mirror reflectivities. These trends are qualitatively correctly predicted by the model.

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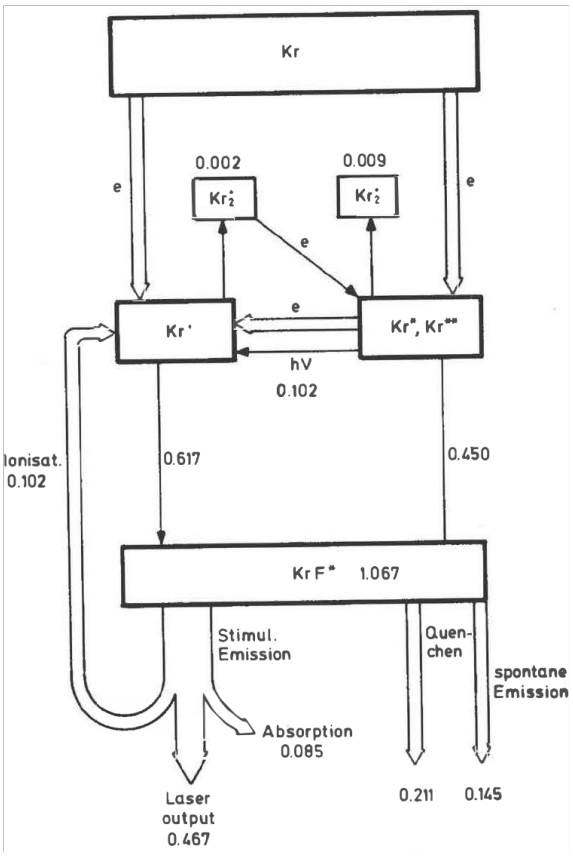


Figure 17: Reaction flows in the KrF laser. The values given refer to the total number of KrF* molecules formed, which is set at 100%. The numbers in parentheses are particle densities in units of 10^{16} cm^{-3} (choice of parameters: losses per resonator pass 20%, $\tau_{\text{e-He}} = 0.3$ ns).

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